Tamika Brown

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Education

University of Waterloo Waterloo, ON
Bachelor of Applied Science, Mechatronics Engineering *Sept 2025 – May 2030*

Experience

Operations & Field Engineering Co-op Student Dec 2025 – May 2026
West Coast Robotics *Agassiz, BC*

- Led field root-cause analysis for Lely Discovery 120 crashes, identifying manure buildup on optical sensors; designed and prototyped brush brackets using hydraulic pipe benders and grinders, culminating in a SolidWorks CAD manufacturing design.
- Built and maintained Power BI analytics dashboards to diagnose and monitor robot crashes, capturing crash type, time, IDs, and weekly counts by farm to support data-driven service planning.
- Supported the installation of two Lely A5 milking robots; wired Ethernet networks, connected hydraulic lines, and configured machinery using ferrule crimpers and a Sawzall, completing tasks ahead of schedule.
- Assisted in deploying two 360 Rain irrigation systems, base stations, and tractor planter kits; conducted land surveys, uploaded planter data via SIM card, and resolved hardware compatibility issues by adding custom tractor hitches.
- Developed an Overstock/Deadstock Power BI report identifying $1.7M in overstock and $1.2M in deadstock; created a regional customer map filtered by robot model to streamline technician service-route planning.

Captain & Driver Sept 2022 – Jun 2025
FIRST Robotics Club 9465 Imagine Axolotls *Chilliwack, BC*

- Assembled and wired chassis and motor controllers from technical diagrams; conducted bench tests and on-robot validations to verify wiring integrity, safety, and performance reliability.
- Built and maintained a climbing arm mechanism with a 75% success rate; performed rapid mechanical and electrical repairs between qualification matches to maximize competitive uptime.
- Led a 9-member FIRST Pacific Regionals team; commanded robot operations across 11 high-stakes matches and coordinated pit redesigns to address sudden performance and maintenance constraints.
- Delivered four STEAM career presentations to 200+ students at SFU and UW to promote robotics and healthcare automation; developed engaging technical demos and Q&A sessions to inspire STEM enrollment.

Operations and Dynamics Member May 2026 – Present
University of Waterloo Baja SAE Team *Waterloo, ON*

- Simulated critical off-road load cases using SolidWorks FEA on upper and lower control arms to predict structural failures, analyze factor of safety, and optimize suspension component geometry.
- Assisted in high-impact sponsorship packages and managed strategic corporate outreach, successfully expanding the team's operational budget and securing critical raw manufacturing resources.
- Streamlined team operations and project lifecycle efficiency by co-developing a comprehensive Work Breakdown Structure (WBS) and maintaining dynamic Gantt charts to track multi-disciplinary deadlines.

Projects

Arcade Machine | *C#, Unity, Photoshop, Laser Cutter* Feb 2023 – June 2023

- Designed and fabricated a functional arcade machine from scratch; milled 10+ wooden components using a chop saw, miter saw, jig saw, and router before assembling into a finished custom cabinet.
- Integrated joysticks, buttons, monitor, and PC by designing and soldering custom wiring harnesses, ensuring reliable input response and clean, organized cable management.
- Developed an interactive game in C# and Unity, deploying it to the finished arcade cabinet and engaging 300+ users at competitive school events.

Video Game Development | *C#, Unity, Photoshop* Sept 2021 – June 2024

- Designed and programmed 10+ Unity-based games in C# using modular, data-driven architectures (ScriptableObjects, object pooling) to support side-scrollers, dungeon crawlers, and physics puzzles.
- Built core gameplay systems including inventory UI with drag-and-drop mechanics, item stacking, save/load states, and level progression; collaborated with artists to integrate Photoshop assets and animations.
- Created and integrated visual assets in Photoshop (sprite sheets, textures, UI graphics) and optimized asset pipelines via sprite atlases to ensure smooth 60 FPS gameplay and crisp visuals.

Technical Skills

Software: C#, C++, Python, AutoCAD, Solidworks, Photoshop, Unity, Lightburn, Power BI, Excel
Hardware: RoboRIO, Microbit, mechanism design/assembly, VEX Robotics
Fabrication: Laser Cutter, Saws, Milling Machine, Lathe, Drill Press